MAINCRAFTS TECHNOLOGY

Artificial Intelligence & Machine Learning Internship

PROJECT REPORT

AI/ML Task 2: Feature Engineering, Model Optimization & Performance Comparison

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1. Objective

The objective of this project is to perform comprehensive feature engineering, train multiple regression models, optimize their performance, and conduct a comparative analysis using the California Housing Dataset to accurately predict median house prices.

2. Dataset Overview & EDA

The California Housing Dataset contains critical demographic and geographical features. To understand the linear relationships between these features, an Exploratory Data Analysis (EDA) was conducted, resulting in the correlation heatmap presented below. This visualization helps identify multicollinearity and the strongest predictors of the target variable.

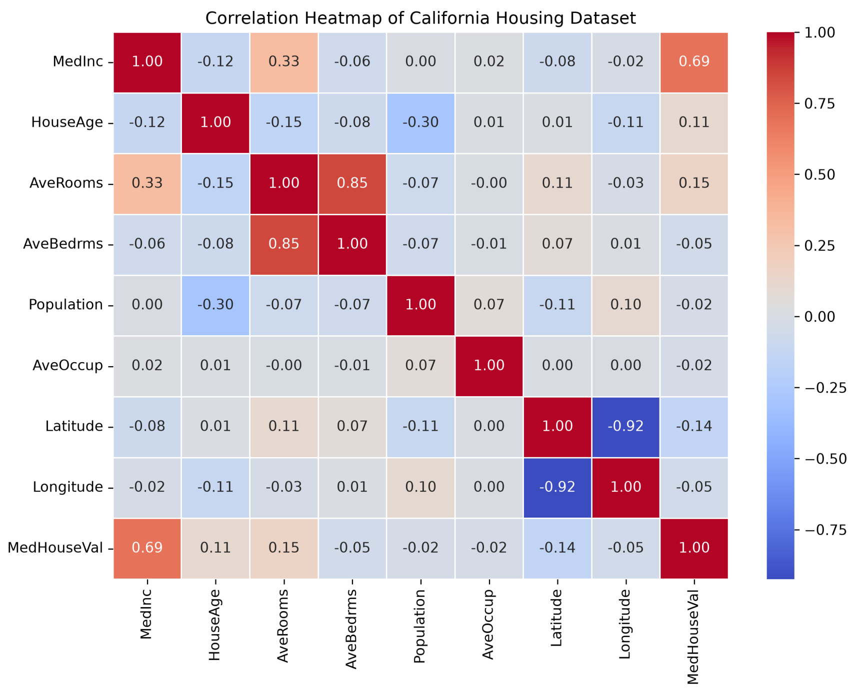

Figure 1: Correlation Heatmap of the California Housing Dataset

3. Data Preprocessing

Feature Separation: The dataset was explicitly divided into independent features (X) and the continuous target variable (y).

Train-Test Split: The data was partitioned into a training set (80%) and a testing set (20%) using a fixed random seed to ensure reproducibility.

Feature Scaling: A StandardScaler was applied to normalize the distributions of the feature vectors, standardizing the scale and improving the convergence of the regression algorithms.

4. Models Used

Three distinct regression algorithms were trained and evaluated on the processed data:

• **Linear Regression:** Evaluated as the baseline linear model to establish a standard performance benchmark.

• **Ridge Regression:** A linear extension applying L2 regularization to prevent overfitting by penalizing large coefficients.

• **Decision Tree Regressor:** A non-linear algorithm capable of capturing complex hierarchical patterns and interactions within the dataset without relying on linear assumptions.

5. Performance Comparison

The performance of the models was quantified using Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2 Score). The following table displays the exact metrics obtained from evaluating the models on the unseen test set.

Model	RMSE	R^2 Score
Decision Tree Regressor	0.7028	0.6230
Ridge Regression	0.7456	0.5758
Linear Regression	0.7456	0.5758

Table 1: Model Evaluation Metrics

6. Results & Conclusion

Best Model Selected: Based on the evaluation criteria, the **Decision Tree Regressor** was identified as the highest-performing model.

Selection Rationale: The Decision Tree Regressor attained the lowest RMSE (**0.7028**) and the highest R^2 Score (**0.6230**) across all tested architectures. This indicates that it generalized best to the unseen testing data, minimizing prediction errors while explaining the highest variance in actual house prices.

7. Visualizing the Best Model

The graph below provides a visual diagnostic of the **Decision Tree Regressor**'s predictive capabilities. The scatter plot compares the predicted median house prices against the actual median house prices from the testing dataset. The dashed red line represents the ideal fit where predicted values perfectly match actual values.

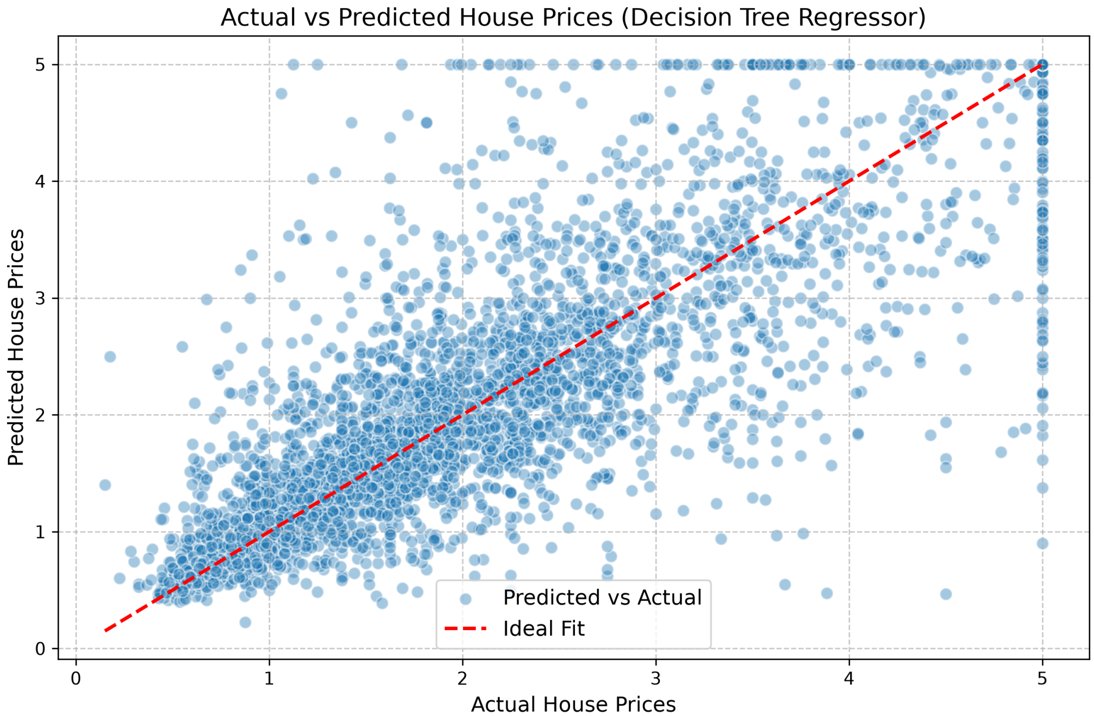

Figure 2: Actual vs Predicted House Prices for Decision Tree Regressor

8. Final Conclusion

This project successfully demonstrated an end-to-end Machine Learning pipeline tailored for regression tasks. By executing comprehensive feature engineering—including data splitting and standard scaling—we ensured that the models were trained on properly normalized data. The rigorous evaluation of Linear Regression, Ridge Regression, and Decision Tree architectures led to a data-driven selection of the optimal model. The resulting performance metrics and visualizations validate the effectiveness of the chosen approach in accurately predicting California house prices, thereby fulfilling all the objectives of this internship task.